

Pregame Pitcher Strikeout Projection — Project Summary

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Companion to the full technical manuscript in `docs/paper/manuscript.md`.

Goal

Estimate a starter’s strikeout rate before first pitch, project how many batters that starter will face, and convert the pair into expected strikeouts and $P(K \geq L)$ for common prop lines L —using only information available before first pitch:

$$k_{\text{rate}} \times \text{TBF} \rightarrow E[K] \rightarrow P(K \geq L)$$

Models and headline metrics

Component	Model	Role
Strikeout rate	Unweighted LightGBM (184 features)	Frozen production rate model
Batters faced (TBF)	Ridge (thin bullpen, 24 features)	Projected exposure
Counts / lines	Binomial / Poisson on projected TBF	Expected K and $P(K \geq L)$

Frozen rate model (2023–2024 chronological test): MAE / RMSE / $R^2 \approx \mathbf{0.0787} / \mathbf{0.0987} / \mathbf{0.147}$. Same test: Marcel-lite MAE $\approx \mathbf{0.0826}$; train-mean MAE $\approx \mathbf{0.0854}$.

Count layer (projected-TBF exposure): expected-K MAE / RMSE / $R^2 \approx \mathbf{1.790} / \mathbf{2.213} / \mathbf{0.168}$; line Briers roughly **0.12–0.22** (lines 3.5–7.5).

Full-stack evaluation

Evaluation gate	Result
Estimator tuning	Keep baseline LightGBM defaults; Ridge α tuned and persisted
Walk-forward stack backtest	Mean expected-K MAE $\approx \mathbf{1.778}$ (3 expanding 2024 windows; $\sigma \approx 0.036$)
Calibration	Mean ECE $\approx \mathbf{0.024}$ (11.C diagnose); production Platt post-hoc on <code>p_over_*</code> (chrono CV $\Delta\text{ECE} \approx -0.008$; raw retained). Fit on 2024 walk-forward OOS predictions; not yet refit on 2026 conditions.
Population policy	Metrics conditional on $\text{PA} \geq 9$ (~3.5% of first pitchers excluded)
Live market pilot (exploratory)	Paper-traded vs. real DK/FD lines, quarter-Kelly sizing off an 8% edge floor. Pre-registered gate: $n \geq 100$ CLV samples + bootstrap CI excluding 0. Current status: inconclusive — directionally positive mean CLV ($\sim +0.7\text{pp}$) at $n \approx 85\text{--}90$, below the pre-registered sample threshold. Not a claim of market edge.

Takeaway

Leakage-safe Statcast $\rightarrow$ rate $\times$ exposure stack with nested chronological selection. Clears a Marcel-lite talent floor on chronological test; opponent lineup is the only leave-family-out family with both-fold within-fold bootstrap support. Absolute R^2 remains limited (~ 0.15). A live closing-line-value pilot against real sportsbook markets is running under a pre-registered statistical gate — currently inconclusive, to be updated once the sample resolves. Portfolio claim: ML engineering under a hard pregame constraint, extended with production deployment and live-market monitoring discipline.

Full write-up: [docs/paper/manuscript.pdf](#)